

Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

L-3/T-2

July 2025

CSE 311: Data Communication

Time: 25 minutes**Marks: 20**

Student Name: _____

Student ID: _____

1. Consider, you are sending some binary data from one device to another. (10)
 - (a) Suppose, you have data elements 010010. Now, convert them to signal elements according to the following schemes: (3+3)
 - i. Bipolar AMI
 - ii. Bipolar PseudoternaryUse vertical dashed bar to differentiate each bit while drawing.
 - (b) Now, you have to send a long sequence of 0's. Is Bipolar AMI suitable for this scenario? If not, what specific problems does it exhibit? What technique(s) can be used to mitigate these issues? Briefly explain the technique(s). (4)

2. Consider a message signal whose amplitude is 128 units and band is limited 5 KHz. It is sampled using Nyquist Rate i.e. the minimum sampling frequency. (10)

- (a) Suppose, you are using Pulse Code Modulation(PCM). If the amplitude range is divided into L sub-intervals and step size is 4 units, what is the Data rate for this signal? You can consider that max value of the range is same as the amplitude of message signal. (5)

Hint: Try to find how many binary bits are needed for the quantization level. Then Nyquist Rate will help you.

- (b) Now, you are using Delta Modulation(DM) instead of PCM. But you are facing Slope Overhead. Explain why or when this problem occurs in DM? How can you overcome this problem? Is there any trade-off if we implement the solution? If yes, what is the best agreement to win the tradeoff? (5)